

Capture the Flag 2: Widgets

Goal: Turn the single-card application into a simple multi-section interface.

Note from the instructor before you start:

These activities involve using your own problem solving and researching skills. We are strictly to implement NOT TO USE AI to generate the code for you. You may only use AI for RESEARCHING the syntax or usage, but NEVER THE CODE ITSELF.

Consequence: Immediate 0 score in your grade for this activity.

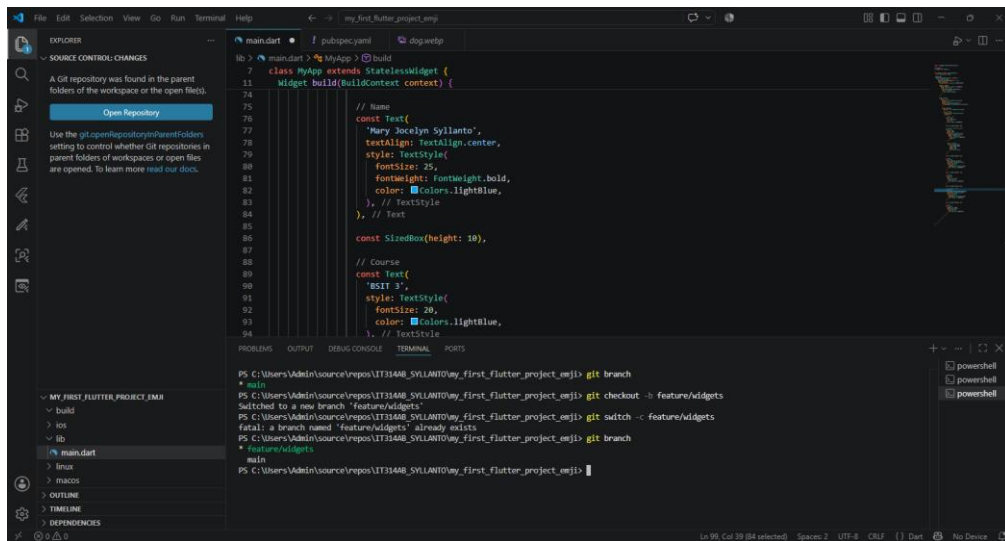
(I know what AI-generated code looks like, don't even think about it :>)

🚩 Flag 0 – Create Your Feature Branch

Objective: Create a feature branch before modifying your application. Name the branch feature/widgets

Screenshot:

- Terminal showing `feature/widgets`



🚩 Flag 1 – Break It Apart

Objective: Separate your profile information into individual **Text** widgets.

If your card currently looks like this:

```
Text(  
  'Chiara Marie C. Canque\n'  
  'BSIT - 3\n'  
  'My First Flutter Application\n'  
  'August 11, 2026',  
)
```

Your task: Separate each piece of information into its own **Text** widget.

You should have something conceptually like:

Text → Name

Text → Course & Section

Text → Application Title

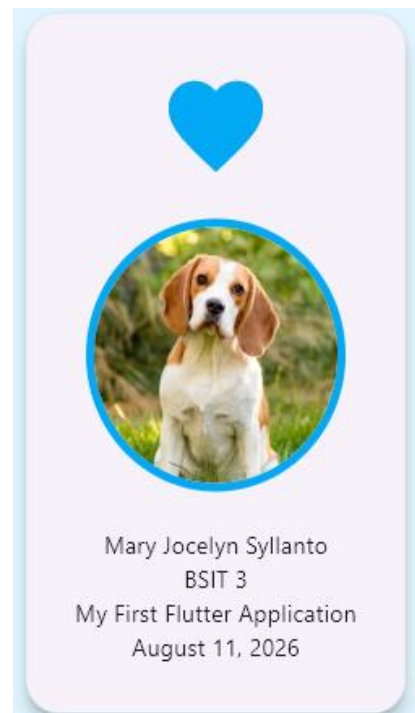
Text → Date

Do not worry about making it pretty yet.

Screenshot:

- Your code showing the separate `Text` widgets
- Your application

```
Column(children: [  
  Text('Mary Jocelyn Syllanto'),  
  Text('BSIT 3'),  
  Text('My First Flutter Application'),  
  Text('August 11, 2026'),  
]), // Column
```



🚩 Flag 2 – Stack the Information

Objective: Arrange your individual `Text` widgets vertically.

Research how to use:

`Column()`

Place your information inside the `Column`.

Your card should now have:

Name

Course & Section

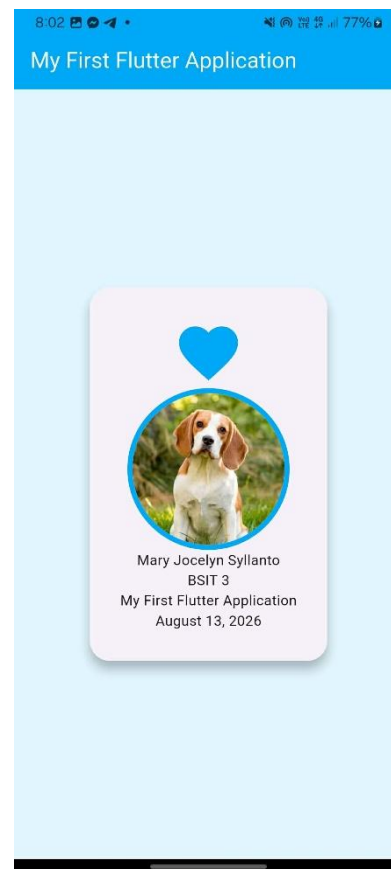
Application Title

Date

Screenshot:

- The `Column` in your code
- The four separate pieces of information displayed vertically

```
38 child: Column(  
39   mainAxisAlignment: MainAxisAlignment.min,  
40   children: [  
41  
42     // Heart icon  
43     const Icon(  
44       Icons.favorite,  
45       size: 70,  
46       color: Colors.lightBlue,  
47     ),  
48  
49     const SizedBox(height: 20),  
50  
51     // Circular profile picture  
52     Container(  
53       width: 160,  
54       height: 160,  
55       decoration: BoxDecoration(  
56         shape: BoxShape.circle,  
57         border: Border.all(  
58           color: Colors.lightBlue,  
59           width: 5,  
60         ),  
61       ),  
62  
63       child: ClipOval(  
64         child: Image.asset(  
65           'assets/dog.webp',  
66           width: 150,  
67           height: 150,  
68           fit: BoxFit.cover,  
69         ),  
70       ),  
71     ),  
72  
73     const SizedBox(height: 20),  
74  
75     // Name  
76     Text(  
77       'Mary Jocelyn Syllanto',  
78     ),  
79  
80     // FLAG 1: Course & Section  
81     Text(  
82       'BSIT 3',  
83     ),  
84  
85     // FLAG 1: Application Title  
86     Text(  
87       'My First Flutter Application',  
88     ),  
89  
90     // FLAG 1: Date  
91     Text(  
92       'August 13, 2026',  
93     ),  
94   ],  
95 ),
```



Flag 3 – Add Another Element

Objective: Add another individual widget to your **Column**.

Add something that belongs to your profile.

Examples:

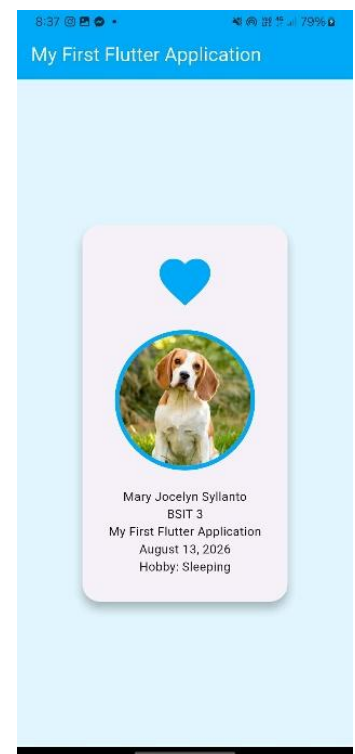
Hobby

It must be its **own widget**, not additional text inside another **Text**.

Screenshot:

- New widget in your **Column**
- Updated application

```
72  
73     const SizedBox(height: 20),  
74  
75     // Name  
76     Text(  
77       'Mary Jocelyn Syllanto',  
78     ),  
79  
80     // FLAG 1: Course & Section  
81     Text(  
82       'BSIT 3',  
83     ),  
84  
85     // FLAG 1: Application Title  
86     Text(  
87       'My First Flutter Application',  
88     ),  
89  
90     // FLAG 1: Date  
91     Text(  
92       'August 13, 2026',  
93     ),  
94  
95     Text(  
96       'Hobby: Sleeping'  
97     ),  
98   ],  
99 ),  
100 ),
```



🚩 Flag 4 – Put Two Things Side-by-Side

Objective: Discover the **Row** widget.

Create a new section containing **two pieces of information beside each other**.

For example:

Age: 21 Birthdate: August 11, 2003

Research:

Row()

Place the Row inside your existing Column.

Your structure should now resemble:

Column

└─ Text

└─ Text

└─ Text

└─ Text

└─ Text

└─ Row

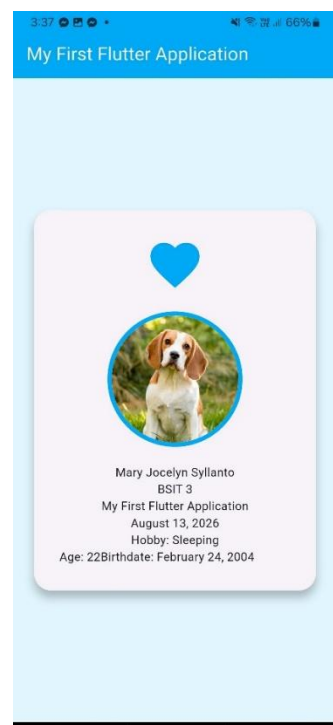
└─ Text

└─ Text

Screenshot:

- Your widget structure/code
- The two items displayed side-by-side

```
Row(  
  children: [  
    Text('Age: 22'),  
    Text('Birthdate: February 24, 2004'),  
  ],  
)
```



🚩 Flag 5 – Give It Some Space

Objective: Improve the spacing between your widgets.

Research:

`SizeBox()`

Add appropriate spacing between your:

- Image
- Name
- Course
- Other information
- Row

Challenge: Make the card easier to read **without changing its colors or fonts yet.**

Screenshot:

- `SizeBox` usage
- Updated application

```
// Name
Text(
  'Mary Jocelyn Syllanto',),
  SizeBox(height: 12), |

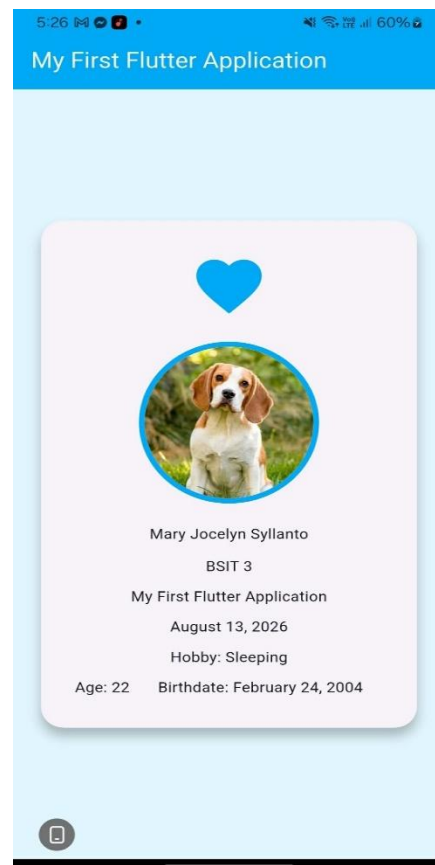
// FLAG 1: Course & Section
Text(
  'BSIT 3',),
  SizeBox(height: 10),

// FLAG 1: Application Title
Text(
  'My First Flutter Application', ),
  SizeBox(height: 10),

// FLAG 1: Birthdate
Text(
  'August 13, 2026',),
  SizeBox(height: 10),

// FLAG 1: Hobby
Text(
  'Hobby: Sleeping',),
  SizeBox(height: 10),

Row(
  children: [
    Text('Age: 22'),
    SizeBox(width: 25),
    Text('Birthdate: February 24, 2004'),
  ],
),
),
),
),
```



🚩 Flag 6 – Make It Look Better

Objective: Apply styling to your individual widgets.

Now that your information is properly separated and positioned, improve the visual hierarchy.

Research how to modify:

`TextStyle()`

Apply different styling to at least **three** elements.

For example:

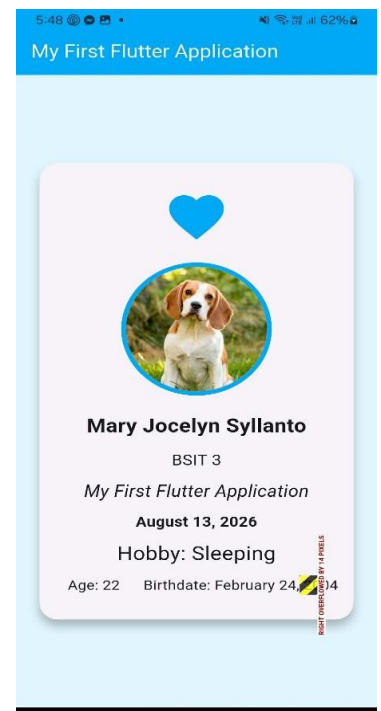
- Name → larger and bold
- Course → medium size
- Date → smaller
- Application title → italic

Screenshot:

- Your `TextStyle` changes
- Updated application

```
75 // Name
76 Text(
77   'Mary Jocelyn Syllanto',
78   style: TextStyle(
79     fontSize: 23,
80     fontWeight: FontWeight.bold,
81   ),
82 ),
83 SizedBox(height: 12,),
84
85 // FLAG 1: Course & Section
86 Text(
87   'BSIT 3',
88   style: TextStyle(
89     fontSize: 18,
90   ),
91 ),
92 SizedBox(height: 10),
93
94 // FLAG 1: Application Title
95 Text(
96   'My First Flutter Application',
97   style: TextStyle(
98     fontSize: 20,
99     fontStyle: FontStyle.italic
100  ),
101 ),
102 SizedBox(height: 10),
103
104 // FLAG 1: Date
105 Text(
106   'August 13, 2026',
107   style: TextStyle(
108     fontSize: 17,
109     fontWeight: FontWeight.bold,
110   ),
111 )
```

```
112 ),
113 SizedBox(height: 10),
114
115 Text(
116   'Hobby: Sleeping',
117   style: TextStyle(
118     fontSize: 23,
119   ),
120 ),
121 SizedBox(height: 10),
122
123 Row(
124   children: [
125     Text('Age: 22',
126       style: TextStyle(
127         fontSize: 16,
128       ),
129     ),
130     SizedBox(width: 25),
131     Text('Birthdate: February 24, 2004',
132       style: TextStyle(
133         fontSize: 16,
134       ),
135     ),
136   ],
137 ),
138 ],
139 ),
140 ),
141 ),
142 ),
143 ),
144 );
145 }
146 }
```



🚩 Flag 7 – Improve the Layout

Objective: Control the alignment of your widgets.

Research the properties of `Column` and `Row`.

Experiment with:

`mainAxisAlignment`

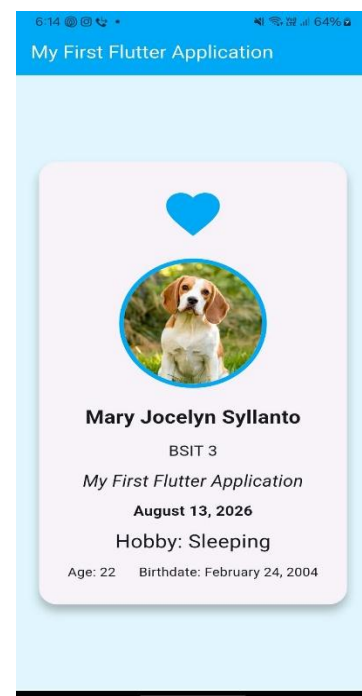
`crossAxisAlignment`

Change the alignment of your content so the card looks intentional rather than simply stacked.

Screenshot:

- The properties you changed
- Updated application

```
123 | Row(  
124 |   mainAxisAlignment: MainAxisAlignment.center,  
125 |   crossAxisAlignment: CrossAxisAlignment.center,  
126 |   children: [  
127 |     Text('Age: 22',  
128 |       style: TextStyle(  
129 |         fontSize: 16,  
130 |       ),  
131 |     ),  
132 |     SizedBox(width: 25),  
133 |     Text('Birthdate: February 24, 2004',  
134 |       style: TextStyle(  
135 |         fontSize: 16,  
136 |       ),  
137 |     ),  
138 |   ],  
139 | ),  
140 | ],
```



🚩 Flag 8 – Create a Second Card

Objective: Apply what you learned to create another section.

Create a second **Card** containing something different from your profile.

For example:

My Favorites

Favorite Game

Favorite Food

Favorite Movie

The second card should also use:

- Individual **Text** widgets
- A **Column**
- At least one **Row**
- Spacing

Screenshot:

- Code for the second card
- Both cards displayed

```
155 // CARD 2: MY FAVORITES
156 Card(
157   color: Colors.lightBlue.withOpacity(0.3),
158   elevation: 0,
159
160   child: Padding(
161     padding: EdgeInsets.all(16),
162
163     child: Column(
164       children: [
165
166         // Favorite Icon
167         Icon(
168           Icons.favorite,
169           size: 25,
170           // Icon
171
172         SizedBox(height: 10),
173
174         // My Favorites
175         Text(
176           'My Favorites',
177           style: TextStyle(
178             fontSize: 22,
179             fontWeight: FontWeight.bold,
180           ), // TextStyle
181         ), // Text
182
183         SizedBox(height: 10),
184
185         // Game and Movie
186         Row(
187           mainAxisAlignment: MainAxisAlignment.center,
188           children: [
189             Text(
190               'Game: Valorant',
```

```
189       Text(
190         'Game: Valorant',
191         style: TextStyle(
192           fontSize: 16,
193         ), // TextStyle
194       ), // Text
195
196       SizedBox(width: 15),
197
198       Text(
199         'Movie: Spiderman',
200         style: TextStyle(
201           fontSize: 16,
202         ), // TextStyle
203       ), // Text
204     ],
205   ), // Row
206
207   SizedBox(height: 10),
208
209   // Color
210   Text(
211     'Color: Black',
212     style: TextStyle(
213       fontSize: 15,
214     ), // TextStyle
215   ), // Text
216 ],
217 ), // Column
218 ), // Padding
219 ), // Card
220
221 SizedBox(height: 20),
222 ],
223 ), // Column
224 ), // Padding
```



🚩 Flag 9 – Stack the Cards

Objective: Arrange your two cards vertically.

You now have:

Profile Card



Favorites Card

Research how to place **multiple widgets inside the Flutter body**.

Use a layout widget to arrange your two cards vertically.

Screenshot:

- The widget containing both Cards
- Both cards visible in the application

```

29 body: SingleChildScrollView(
30   child: Padding(
31     padding: EdgeInsets.symmetric(horizontal: 22),
32
33     // Column arranges the two Cards vertically
34     child: Column(
35       mainAxisAlignment: MainAxisAlignment.center,
36       crossAxisAlignment: CrossAxisAlignment.stretch,
37       children: [
38
39         // CARD 1: PROFILE
40         Card(
41           color: Colors.lightBlue.withOpacity(0.3),
42           elevation: 0,
43
44           child: Padding(
45             padding: EdgeInsets.fromLTRB(
46               16,
47               16,
48               16,
49               0,
50             ), // EdgeInsets.fromLTRB
51
52             child: Column(
53               mainAxisAlignment: MainAxisAlignment.center,
54               children: [
55
56                 // Circular Avatar
57                 Container(
58                   width: 160,
59                   height: 160,
60                   decoration: BoxDecoration(
61                     shape: BoxShape.circle,
62                     border: Border.all(
63                       color: Colors.lightBlue,

```

```

99       'My First Flutter Application',
100       style: TextStyle(
101         fontSize: 18,
102         fontStyle: FontStyle.italic,
103       ), // TextStyle
104     ), // Text
105
106     SizedBox(height: 10),
107
108     Text(
109       'August 17, 2026',
110       style: TextStyle(
111         fontSize: 18,
112         fontWeight: FontWeight.bold,
113       ), // TextStyle
114     ), // Text
115
116     SizedBox(height: 10),
117
118     Text(
119       'Hobby: Sleeping',
120       style: TextStyle(
121         fontSize: 15,
122       ), // TextStyle
123     ), // Text
124
125     SizedBox(height: 10),
126
127     Row(
128       mainAxisAlignment: MainAxisAlignment.center,
129       children: [
130         Text(
131           'Age: 22',
132           style: TextStyle(
133             fontSize: 15,

```

```

167     Icons.favorite,
168     size: 25,
169   ), // Icon
170
171   SizedBox(height: 10),
172
173   // My Favorites
174   Text(
175     'My Favorites',
176     style: TextStyle(
177       fontSize: 22,
178       fontWeight: FontWeight.bold,
179     ), // TextStyle
180   ), // Text
181
182   SizedBox(height: 10),
183
184   // Game and Movie
185   Row(
186     mainAxisAlignment: MainAxisAlignment.center,
187     children: [
188       Text(
189         'Game: Valorant',
190         style: TextStyle(
191           fontSize: 16,
192         ), // TextStyle
193       ), // Text
194
195       SizedBox(width: 15),
196
197       Text(
198         'Movie: Spiderman',
199         style: TextStyle(
200           fontSize: 16,
201         ), // TextStyle
202       ), // Text

```

```

65     width: 5,
66   ), // BoxDecoration
67   child: ClipOval(
68     child: Image.asset(
69       'assets/dog.webp',
70       width: 150,
71       height: 150,
72       fit: BoxFit.cover,
73     ), // Image.asset
74   ), // ClipOval
75   ), // Container
76
77   SizedBox(height: 10),
78
79   Text(
80     'Mary Jocelyn Syllanto',
81     style: TextStyle(
82       fontSize: 23,
83       fontWeight: FontWeight.bold,
84     ), // TextStyle
85   ), // Text
86
87   SizedBox(height: 10),
88
89   Text(
90     'BSIT 3',
91     style: TextStyle(
92       fontSize: 18,
93     ), // TextStyle
94   ), // Text
95
96   SizedBox(height: 10),
97
98   Text(
99     'My First Flutter Application',

```

```

130   Text(
131     'Age: 22',
132     style: TextStyle(
133       fontSize: 15,
134     ), // TextStyle
135   ), // Text
136
137   SizedBox(width: 25),
138
139   Text(
140     'Birthdate: February 24, 2004',
141     style: TextStyle(
142       fontSize: 15,
143     ), // TextStyle
144   ), // Text
145
146   ), // Row
147
148   ), // Column
149   ), // Padding
150   ), // Card
151
152   SizedBox(height: 20),
153
154   // CARD 2: MY FAVORITES
155   Card(
156     color: Colors.lightBlue.withOpacity(0.3),
157     elevation: 0,
158
159     child: Padding(
160       padding: EdgeInsets.all(16),
161
162       child: Column(
163         children: [

```

```

193   ), // Text
194
195   SizedBox(width: 15),
196
197   Text(
198     'Movie: Spiderman',
199     style: TextStyle(
200       fontSize: 16,
201     ), // TextStyle
202   ), // Text
203
204   ), // Row
205
206   SizedBox(height: 10),
207
208   // Color
209   Text(
210     'Color: Black',
211     style: TextStyle(
212       fontSize: 15,
213     ), // TextStyle
214   ), // Text
215
216   ), // Column
217   ), // Padding
218   ), // Card
219
220   SizedBox(height: 20),
221
222   ), // Column
223   ), // Padding
224   ), // SingleChildScrollView
225   ), // Scaffold
226   ); // MaterialApp
227
228

```



🚩 Flag 10 – Widget Discovery

Objective: Discover a Flutter widget independently.

Research the Flutter widget catalog and find **one widget you haven't used yet**.

Examples:

- Icon
- Divider
- CircleAvatar
- Chip
- ElevatedButton

Add it somewhere appropriate in your application.

Screenshot:

- Your newly discovered widget in the code
- The widget visible in the application

```
216 | Chip(  
217 |   label: Text(  
218 |     'Color: Black',  
219 |     style: TextStyle(  
220 |       fontSize: 15,  
221 |     ), // TextStyle  
222 |   ), // Text  
223 | ), // Chip  
224 |
```



🚩 Flag 11 – Real Phone Test 📱

Objective: Run your completed application on a **physical** Android phone.

Research how to run Flutter on your phone.

Screenshot:

- Your completed application running on the physical phone (screenshot from your actual phone)



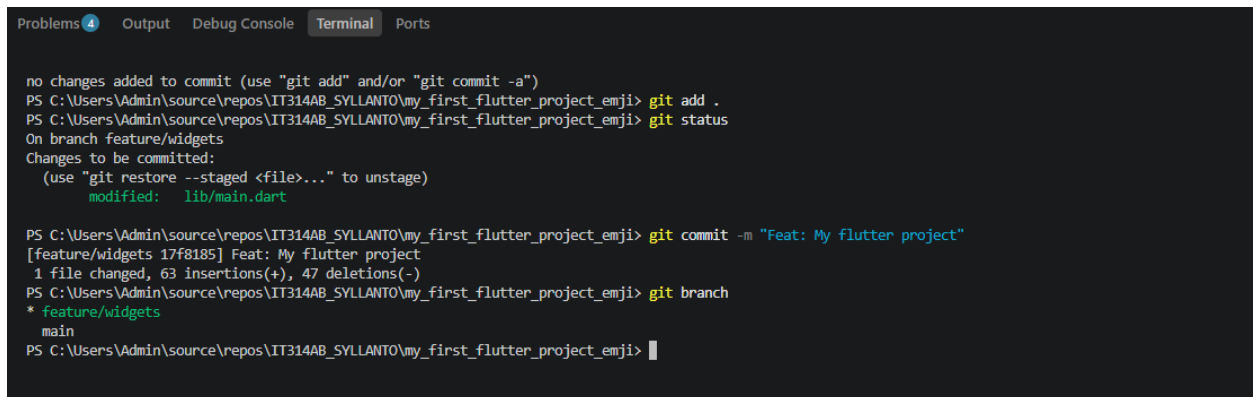
🚩 Flag 12 – Save Your Progress

Objective: Commit your completed Widget branch.

Follow proper standards of commit messages

Screenshot:

- Successful commit
- `feature/widgets` branch



```
Problems 4 Output Debug Console Terminal Ports

no changes added to commit (use "git add" and/or "git commit -a")
PS C:\Users\Admin\source\repos\IT314AB_SYLLANT0\my_first_flutter_project_emji> git add .
PS C:\Users\Admin\source\repos\IT314AB_SYLLANT0\my_first_flutter_project_emji> git status
On branch feature/widgets
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   lib/main.dart

PS C:\Users\Admin\source\repos\IT314AB_SYLLANT0\my_first_flutter_project_emji> git commit -m "Feat: My flutter project"
[feature/widgets 17f8185] Feat: My flutter project
 1 file changed, 63 insertions(+), 47 deletions(-)
PS C:\Users\Admin\source\repos\IT314AB_SYLLANT0\my_first_flutter_project_emji> git branch
* feature/widgets
  main
PS C:\Users\Admin\source\repos\IT314AB_SYLLANT0\my_first_flutter_project_emji>
```